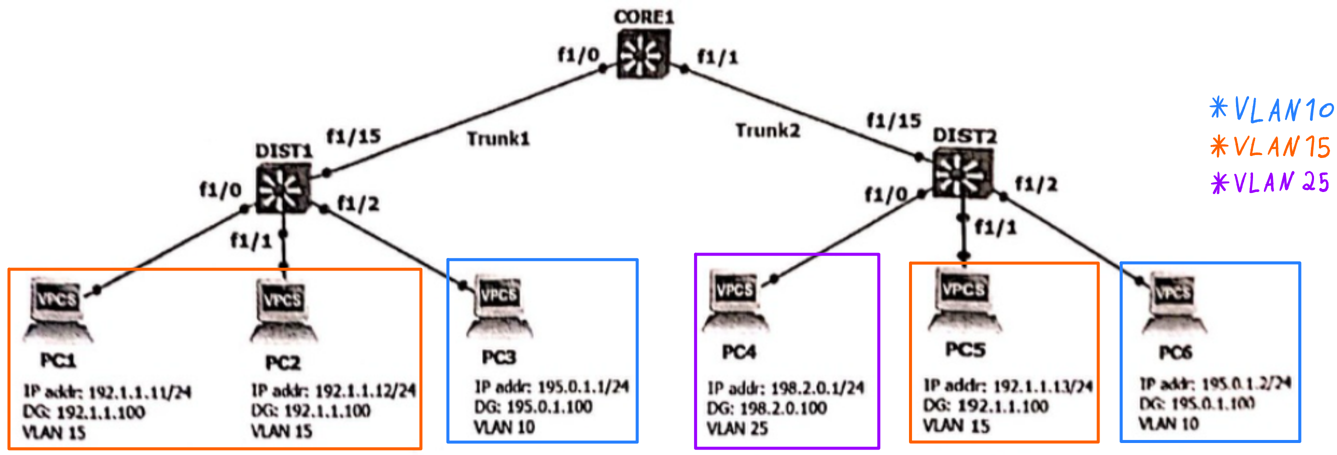
Correção do teste prático

Teste Prático No. 1
Redes de Comunicações II – ano letivo 2024/2025

No. Mec.: _ _ Nome: _ ________

NOTAS: Cada resposta certa vale + 0.625 valores; cada resposta errada vale – 0.3125 valores.

Scenario 1: Consider the following network, where the Default Gateway of the IP networks of VLANs 10 and 15 is **CORE1** and the Default Gateway of the IP network of VLAN 25 is **DIST2**. There is an interconnecting VLAN between **CORE1** and **DIST2**. The trunks support only the VLANs required to provide IP connectivity among all IP networks.



In the IP connectivity between different PCs, the following ICMP packets were captured in **Trunk2**:

PACKET 1 (PC1→PC4)

```
› Frame 56: 102 bytes on wire (816 bits), 102 bytes captured (8
› Ethernet II, Src: c2:01:43:bc:00:00 (c2:01:43:bc:00:00), Dst:
› 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 40
› Internet Protocol Version 4, Src: 192.1.1.11, Dst: 198.2.0.1
⌄ Internet Control Message Protocol
    Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0xd06b [correct]
    [Checksum Status: Good]
    Identifier (BE): 20383 (0x4f9f)
    Identifier (LE): 40783 (0x9f4f)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Response frame: 57]
  › Data (56 bytes)
```

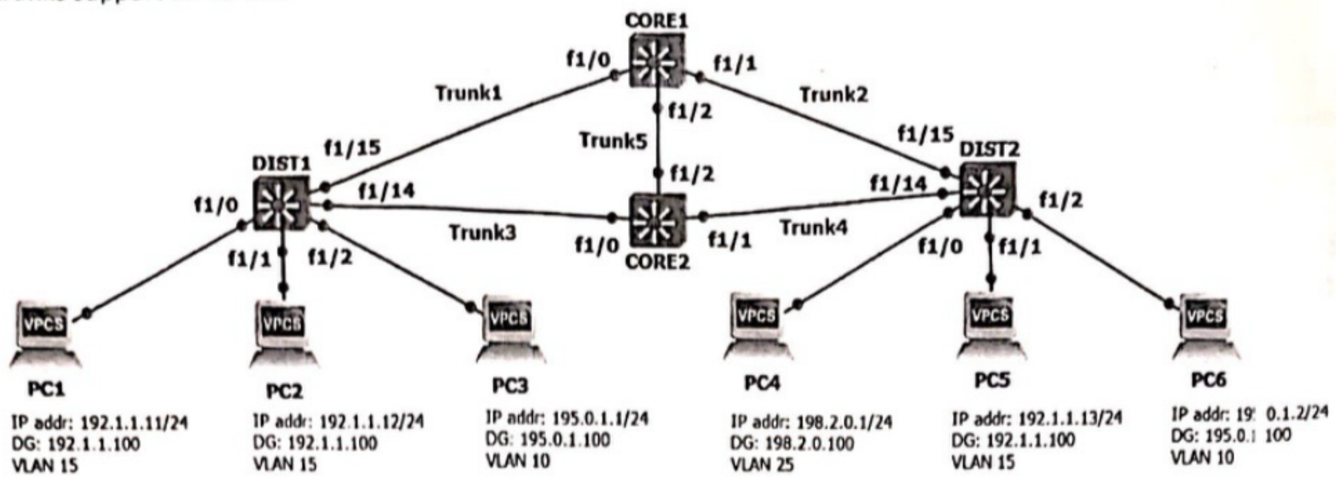
PACKET 2 (VLAN25→PC3)

```
› Frame 799: 102 bytes on wire (816 bits), 102 bytes captured (8
› Ethernet II, Src: c2:03:32:e0:00:00 (c2:03:32:e0:00:00), Dst:
› 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 40
› Internet Protocol Version 4, Src: 200.0.0.20, Dst: 195.0.1.1
⌄ Internet Control Message Protocol
    Type: 0 (Echo (ping) reply)
    Code: 0
    Checksum: 0xa26a [correct]
    [Checksum Status: Good]
    Identifier (BE): 34208 (0x85a0)
    Identifier (LE): 41093 (0xa085)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Request frame: 798]
    [Response time: 15.474 ms]
  › Data (56 bytes)
```

Inside each square, classify as True (T) or False (F) each of the following statements:

- $\boxed{F}$ a) In a ping from PC1 to $\overbrace{\text{192.1.1.12}}^{\text{PC2}}$, ICMP Echo Requests are captured in **Trunk1** in VLAN 15.
- $\boxed{F}$ b) In a ping from PC5 to 198.2.0.23, ARP Requests are captured in **Trunk2** in VLAN 25.
- $\boxed{T}$ c) **Trunk1** and **Trunk2** belong to the broadcast domain of VLAN 10.
- $\boxed{T}$ d) The interconnection VLAN is VLAN 40.
- $\boxed{F}$ e) The IP address 195.0.1.100 belongs to **DIST2**.
- $\boxed{T}$ f) The IP network of the interconnection VLAN can be 200.0.0.0/26.
- $\boxed{T}$ g) PACKET 1 is captured in a ping from PC1 to PC4.
- $\boxed{T}$ h) PACKET 2 is captured in a ping from PC3 to **DIST2**.

Scenario 2: Consider the following network, where the Default Gateway of the IP networks of all VLANs is **CORE1**. All trunks support all VLANs.



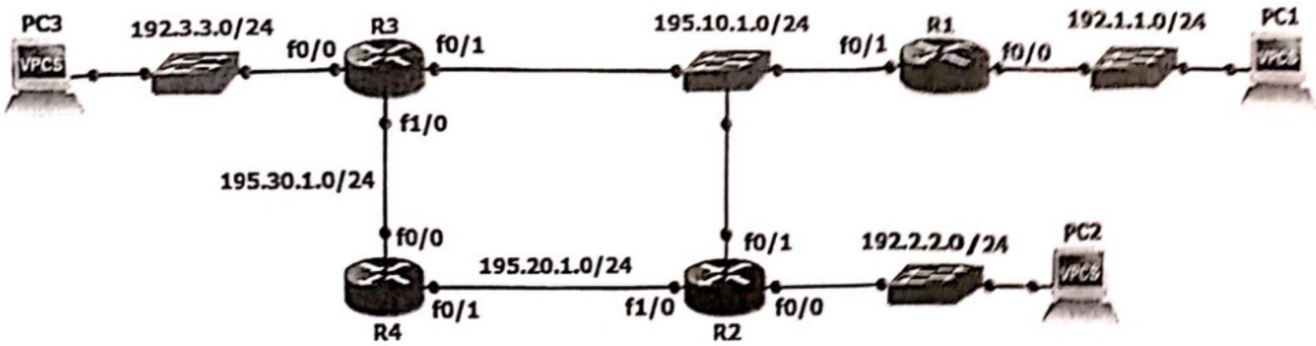
In this scenario, <u>the Root Bridge of VLAN 15 is **DIST1**</u>. The spanning tree brief information of VLAN 15 in **CORE2** is:

```
VLAN15
  Spanning tree enabled protocol ieee
  Root ID    Priority    16384
             Address     c202.19b4.0000
             Cost        19
             Port        41 (FastEthernet1/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32768
             Address     c204.43f8.0002
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time 300

Interface                                  Designated
Name                 Port ID Prio Cost  Sts Cost  Bridge ID            Port ID
-------------------- ------- ---- ----- --- ----- -------------------- -------
FastEthernet1/0      128.41   128    19 FWD     0 16384 c202.19b4.0000 128.55
FastEthernet1/1      128.42   128    19 FWD    19 32768 c204.43f8.0002 128.42
FastEthernet1/2      128.43   128    19 BLK    19 32768 c201.43bc.0002 128.43
```

Inside each square, classify as True (T) or False (F) each of the following statements <u>concerning VLAN 15</u>:

- [ ] a) The Root Bridge has priority 16384.
- [T] b) The port cost of `f1/2` of **CORE2** is 19.
- [F] c) The Designated Bridge of **Trunk1** is **CORE1**.
- [F] d) The interface `f1/0` of **CORE1** is a designated port.
- [T] e) The MAC address of **CORE2** is c2:04:43:f8:00:02.
- [F] f) The interface `f1/14` of **DIST1** is a root port.
- [T] g) Changing the port cost of `f1/0` of **CORE2** to 12 changes the VLAN 15 spanning tree.
- [T] h) Changing the priority of **CORE1** to 24576 does not change the VLAN 15 spanning tree.

Scenario 3: Consider the following IPv4 network, where all routers are configured with the RIPv2 protocol, with split-horizon, and all networks are in the RIP processes of the routers. The host part of the IP addresses in routers is the number of the router name.



In a ping between two of the PCs, the following packet was captured in the network:

```
> Frame 35: 98 bytes on wire (784 bits), 98 bytes captured ('84
> Ethernet II, Src: ca:01:30:54:00:06 (ca:01:30:54:00:06), D t:
> Internet Protocol Version 4, Src: 192.1.1.12, Dst: 192.3.3.'4
v Internet Control Message Protocol
    Type: 0 (Echo (ping) reply)
    Code: 0
    Checksum: 0x01c9 [correct]
    [Checksum Status: Good]
    Identifier (BE): 9791 (0x263f)
    Identifier (LE): 16166 (0x3f26)
    Sequence Number (BE): 4 (0x0004)
    Sequence Number (LE): 1024 (0x0400)
    [Request frame: 34]
    [Response time: 30.823 ms]
  > Data (56 bytes)
```

Inside each square, classify as True (T) or False (F) each of the following statements:

F a) The IP routing table of R4 has the RIP entry:

```
R      192.2.2.0/24 [120/2] via 195.20.1.2, f0/1
```

F b) The IP routing table of R1 has the RIP entry:

```
R      195.2.2.0/24 [120/1] via 195.10.1.2, f0/1
```

T c) The IP routing table of R4 has the RIP entry:

```
R      195.10.1.0/24 [120/1] via 195.20.1.2, f0/1
                     [120/1] via 195.30.1.3, f0/0
```

☐ d) The RIP Response messages sent by R3 through interface `f1/0` include network 195.10.1.0/24 with metric 1.

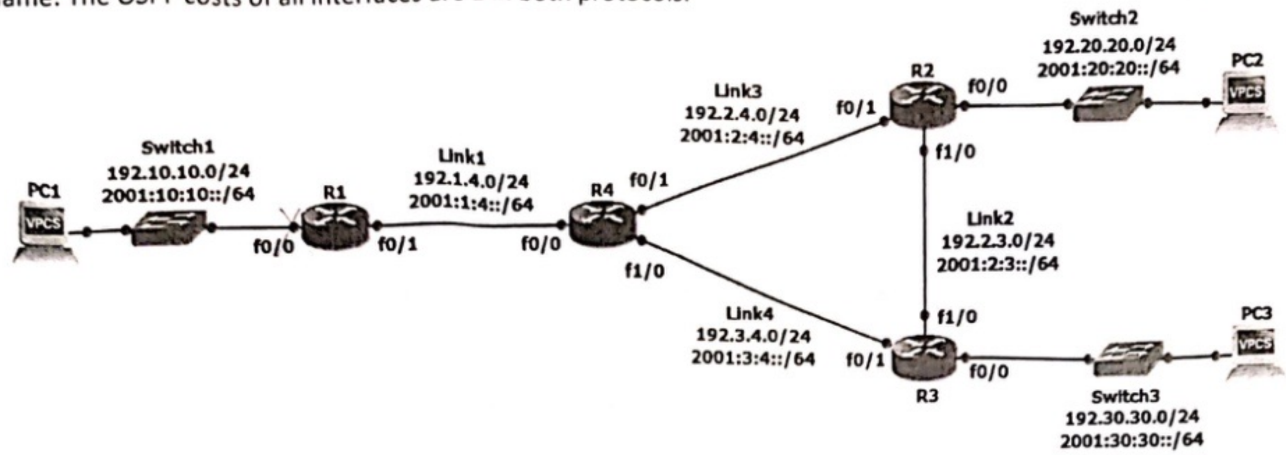
T e) The RIP Response messages sent by R2 through interface `f0/1` contain 2 IP networks.

T f) The IP address of PC1 is 192.1.1.12.

T g) The shown packet was not forwarded for sure through network 195.30.1.0/24.

F h) If interface `f0/1` of R4 is shutdown, the RIP entry of R4 to 192.1.1.0/24 does not change.

Scenario 4: Consider the following dual stack (IPv4 and IPv6) network. All routers are configured with one OSPFv2 process in area 0 and one OSPFv3 process in area 0. In both protocols, all networks except the ones associated with **Switch1** are included in the OSPF processes. The host part of the IP addresses in routers is the number of the router name. The OSPF costs of all interfaces are 1 in both protocols.



The results of command `show ip ospf database` in R4 is:

```
                  Router Link States (Area 0)

Link ID           ADV Router       Age          Seq#        Checksum Link count
192.3.4.4         192.3.4.4        174          0x80000007  0x008136 3
192.10.10.1       192.10.10.1      104          0x80000004  0x00FDF3 1
192.20.20.2       192.20.20.2      250          0x80000006  0x001A12 3
192.30.30.3       192.30.30.3      158          0x80000006  0x00598F 3
                  Net Link States  (Area 0)

Link ID           ADV Router       Age          Seq#        Checksum
192.1.4.4         192.3.4.4        174          0x80000002  0x005BA6
192.2.3.3         192.30.30.3      412          0x80000002  0x009DE7
192.2.4.2         192.20.20.2      250          0x80000002  0x0003CB
192.3.4.3         192.30.30.3      158          0x80000002  0x00059D
                  Type-5 AS External Link States

Link ID           ADV Router       Age          Seq#        Checksum Tag
0.0.0.0           192.10.10.1      104          0x80000002  0x00D506 1
```

Inside each square, classify as True (T) or False (F) each of the following statements:

T a) The IPv4 routing table of R4 has the OSPF entry:

```
O      192.30.30.0/24 [110/2] via 192.3.4.3, f1/0
```

F b) The IPv4 routing table of R2 has the OSPF entry:

```
                         →2
O      192.3.4.0/24 [110/1] via 192.2.4.4, f0/1
                    [110/1] via 192.2.3.3, f1/0
```

F c) If the IPv4 routing table of R3 has the OSPF entry:

```
O*E2   0.0.0.0/0 [110/3] via 192.3.4.4, f0/1
```

it means that the external route of type E2 with an external cost of 2.

T d) The Net Link State ID of **Link2** is 192.2.3.3 in the OSPFv2 domain.

T e) The designated router of **Link4** is R3 in the OSPFv2 domain.

T f) Assuming the link-local addresses are correct, the IPv6 routing table of R3 has the OSPF entry:

```
O   2001:2:4::/64 [110/2]
  via FE80::C802:1EFF:FE6C:1C, f1/0
  via FE80::C804:FFF:FEC0:1C, f0/1
```

T g) The database of the OSPFv3 process has 4 Net Link States.

T h) The network 2001:2:3::/64 is a transit network in the OSPFv3 domain.